University of Asia and the Pacific
School of Sciences and Engineering
Department of Mathematics

# BACHELOR OF SCIENCE IN DATA SCIENCE
# UNDERGRADUATE RESEARCH POLICIES AND GUIDELINES
**(2nd Semester, SY 2024-2025)**

## I. BACKGROUND

The Bachelor of Science in Data Science (BSDS) is a four-year undergraduate program that includes the study of data and the methodologies, processes, algorithms and systems for collecting, refining, storing and analyzing data to arrive at useful insights and knowledge. It is a discipline in computing that benefits from developments in computer science, mathematics, statistics, business and other disciplines. Its graduates are equipped with knowledge and skills for processing, management and analysis of data in various forms. They are competent, innovative, creative and responsible problem-solvers with commitment to contributing to nation-building.

Undergraduate Research is a three-unit course in the BSDS program whose main goal is for students to integrate their learnings in the program. Specifically, students should be able to apply the science of analytics to data to solve a problem in their chosen domain – could be in the field of banking and finance, healthcare, energy, business and economics, telecommunications, or education, to name a few – in order to aid in decision-making or contribute to policy-making, and guided by humanist, ethical, and legal perspectives.

## II. DEFINITION OF TERMS

**Research Project** – the required output of the undergraduate research course. It is an independent individual analytics research work which satisfies the above-mentioned goal and must be presented before a panel.

**Research Student** – a student who is enrolled in the undergraduate research course, which is usually offered in the second semester of the fourth year of the BSDS Program. He/she must be at least of fourth year standing.

**Faculty-in-Charge** – a full-time faculty member who oversees the entire undergraduate research course. He/she determines whether students have submitted acceptable research projects required by the program. He/she may be assisted by a faculty assistant.

**Faculty Assistant** – a full-time or part-time faculty member who assists the faculty-in-charge in any of his/her responsibilities.

**Research Adviser** – a full-time or part-time university employee who works closely with the student in developing a research topic and organizing a plan for investigating it. He/she makes himself/herself available to meet regularly with the student to evaluate the progress of his or her research, to discuss the problems that arise, and to provide the necessary support and direction. He/she is expected to check the manuscript of his/her advisee and give an endorsement on the readiness of the research project for final presentation.

**Research Consultant** – usually an external faculty or a professional who is an expert in a specific domain on which the research project solution is applied. He/she may perform the duties and responsibilities of, though not limited to, a research adviser or a research panelist.

**Research Panelist** – a full-time or part-time employee or a non-UA&P employee who reviews the quality of the research project deliverables and evaluates the student's ability to respond to questions related to the research project.

## III. GENERAL POLICIES

1. As a culminating requirement of the BSDS Program, the student of the undergraduate research course must complete a research project that applies the science of analytics to data to solve a problem in his chosen domain – could be in the field of banking and finance, healthcare, energy, business and economics, telecommunications, or education, to name a few – in order to aid in decision-making or contribute to policy-making, and guided by humanist, ethical, and legal perspectives. The analytics solution covered by the research project should be presented to a panel of evaluators.

2. Through the research project, the student should be able to demonstrate the following:

    a) Design and implement data management processes from acquisition or creation, storage, retrieval to maintenance of the data that will be analyzed
    b) Correctly apply analytics processes, techniques, algorithms and tools on actual data to derive the insights that were deemed to solve a problem or challenge for an organization
    c) Propose actionable recommendations to address the identified problem
    d) Plan, manage, evaluate, and direct the research project from beginning to end

e) Identify, define and resolve ethical and legal concerns specific to the research project as they pertain to persons, organizations and society
f) Follow the research process in completing the project
g) Effectively communicate the research project in oral and written form

3. Students are expected to employ the scientific (or empirical) method of acquiring knowledge from observations. In general, this consists of the following steps:

   a) Defining the problem that needs to be resolved
   b) Finding out as much information (what has been said and done) about the problem
   c) Collecting relevant data
   d) Conducting initial exploration of data to demonstrate feasibility and/or development of a proof of concept
   e) Developing the model given the full dataset
   f) Analyzing the results of the data analytics project
   g) Concluding the project by raising implications of the analysis and proposing actionable recommendations to address the identified problem.

4. The research project is presented before a panel in two parts: (1) project proposal presentation and (2) final project presentation. At the beginning of the semester, each of the three (3) chapters of the research project proposal must be presented in class to get initial feedback from the faculty-in-charge.

## IV. ROLES AND RESPONSIBILITIES

The following are the tasks expected of the following:

**1. Faculty-in-Charge**

- Drafts the course syllabus, plans the course activities and have them approved by the BSDS Program Director
- Orients research project advisers, consultants, and students on the research requirements and deadlines
- Holds classes/seminars/workshops that will aid the students in their research work (for example, give or facilitate talks/workshops on project development and management, showcase outstanding research projects in Data Science particularly of alumni, etc.)
- Gives initial review and approval of the student's research project concept, literature review and methodology
- Receives student deliverables (project proposal and final research project) for panel presentations

- Assigns research project advisers to students
- Convenes the panel for student presentations (project proposal and final research project)
- Addresses any concern of students, advisers, and panelists related to the Undergraduate Research course
- Submits the final grade on time

2. **Research Adviser**

- Makes himself/herself available to meet with the student at least five (5) times, during the semester, to evaluate the progress of the student's research, to discuss the problems that can arise and to provide the necessary support and direction
- Reviews and comments on the student's research project deliverables, including the research manuscript
- Endorses the student's research project deliverables (including the research manuscript) before the student submits to the faculty-in-charge
- Monitors the progress of the student's research project
- Attends the student's research panel presentations (project proposal and final project).

3. **Research Consultant**

- Serves as a resource person to a lecture/seminar/workshop pertaining to a domain identified in the research project proposal
- May also serve as a technical adviser, in partnership with the research adviser. He/she makes himself/herself available to guide and advise the assigned student on his/her research project requirements, especially those that relate to his expertise
- Reviews and comments on the student's research project deliverables

4. **Research Panelist**

- Reviews and comments on the quality of the student's research project deliverables, including the research manuscript
- Attends the student's research project panel presentations to evaluate the student's ability to respond to questions related to the research project deliverables

## V. MECHANICS

1. The initial review and approval of the student's research project concept, literature review and methodology shall be done by the faculty-in-charge through class presentations. The following are the prescribed number of slides for each of the class presentations:

   - **Project Concept Presentation (10 slides):** 1-background, 2-statement of the problem, 3-research objective, 4-solution requirements, 5-significance, 6-scope and limitations, 7-data set source and description, 8-methodology, 9-expected analytics results, and 10-ethical considerations

   - **Literature Review Presentation (10 slides):** based on at least 3 scholarly papers related to the student's research project with summary of methodology and findings

   - **Methodology Presentation (10 slides):** explanation of methods regarding data exploration and algorithms to be used; should also include a plan for the entire research project

2. The Research Project Proposal presentation deck will be a compilation of the revised project concept, literature review and methodology.

3. For the presentation of the Research Project Proposal, the faculty-in-charge will assign three (3) panelists for each student. A student will be given a maximum of thirty (30) minutes to present his/her project proposal and another ten (10) minutes to answer questions from the panel.

4. The following are the possible evaluation outcomes for the proposal presentation:

| Overall Evaluation | Description | Range of Grade |
|---|---|---|
| Accept the proposal (with minor issues) | The proposal is already accepted; the student just needs to address very minor issues. | 1.50 – 1.00 |
| Accept the proposal (with major issues) | The proposal is accepted, in principle but the student has to address some major issues identified by the panel. | 2.25 – 1.75 |
| For further review | The proposal is not yet accepted; the student can still pursue the same topic after addressing the major issues identified by the panel. | 2.50 – 3.00 |
| Reject | The proposal has serious shortcomings; the student has to come up with another proposal. | INC or 3.5 |

5. The faculty-in-charge releases to the student the panel's consolidated comments one (1) week after the proposal presentation. These comments should be incorporated in the written proposal, which is a paper composed of three (3) chapters: Introduction (project concept), Review of Related Works and Methodology.

6. After a successful research project proposal presentation, the student will be formally assigned a research adviser and panelists for the final project presentation.

7. The student then works with his/her adviser in completing the research project, following his/her project plan.

8. A colloquium is scheduled in the middle of the semester as a venue for students to present updates of their research projects. This is open to anyone who is interested in attending.

9. The final research paper consists of the following parts:

   Chapter 1: Introduction
   Chapter 2: Review of Related Works
   Chapter 3: Methodology
   Chapter 4: Data Understanding
   Chapter 5: Data Preparation
   Chapter 6: Modeling
   Chapter 7: Evaluation
   Chapter 8: Results and Discussion
   Chapter 9: Conclusions
   Chapter 10: Recommendations

10. The final research paper should be initially submitted by the student to the panel for review at least one (1) week before the final project presentation, after having obtained the endorsement of his/her adviser.

11. In principle, the composition of the panel for the proposal presentation is the same as the final project presentation.

12. The following are the possible evaluation outcomes for the final research project presentation:

| **Overall Evaluation** | **Description** | **Range of Grade** |
|---|---|---|
| Accept the final paper (with minor issues) | The final paper is already accepted; the student just needs to address very minor issues. | 1.50 – 1.00 |
| Accept the final paper (with major issues) | The final paper is accepted, in principle but the student has to address some major issues identified by the panel. | 2.25 – 1.75 |
| For further review | The final paper is not yet accepted; the student can still pursue the same topic after addressing the major issues identified by the panel. | 2.50 – 3.00 |
| Reject | The final paper has serious shortcomings; the student has to come up with another paper of the same topic or come up with a new research proposal, which will require panel another panel presentation and approval. | INC or 3.5 |

13. Again, the faculty-in-charge releases to the student the panel's consolidated comments a week after the final research project presentation. These comments should be incorporated in the written final paper, which will be the basis for its final rating.

14. The following is the grading system for the Undergraduate Research Project
    - Project Concept Presentation – 10%
    - Literature Review Presentation – 10%
    - Methodology Presentation – 10%
    - Project Proposal Presentation – 10%
    - Written Research Project Proposal – 10%
    - Colloquium – 10%
    - Final Research Project Presentation – 20%
    - Final Research Paper – 20%

## VI. SCHEDULE OF ACTIVITIES/DELIVERABLES

| Date | Meeting | Activities/Deliverables |
|---|---|---|
| Jan 18 | 1 | Introduction to the Undergraduate Research Course |
| Jan 25 | 2 | Research Break |
| Feb 1 | 3 | *Class Presentation: Project Concept* |
| Feb 8 | 4 | *Class Presentation: Literature Review* |
| Feb 15 | 5 | *Class Presentation: Methodology* |
| Feb 22 | 6 | Research break |
| Mar 1 | 7 | *Panel Presentation: Research Project Proposal* |
| Mar 8 | 8 | Panel releases consolidated comments for students |
| | | Research break |
| Mar 15 | 9 | *Submission: Written Research Project Proposal* |
| | | Workshop: Business Communications and Storytelling |
| Mar 22 | 10 | Workshop: Research Writing |
| Mar 29 | 11 | Research Break |
| Apr 5 | 12 | *Colloquium: Research Project Updates* |
| Apr 12 | 13 | Seminar: Alumni Research Projects |
| Apr 19 | 14 | Black Saturday |
| Apr 26 | 15 | *Initial Submission: Final Research Paper for Panel Review* |
| May 3 | 16 | *Panel Presentation: Final Research Project* |
| May 10 | 17 | Panel releases consolidated comments for students |
| May 17 | 18 | *Final Submission: Final Research Paper* |